

Tool reference

AE MCP Bridge 0.2.0: 34 tools. Generated from `src/mcp-server.mjs` by `scripts/gen-tools-doc.mjs` ; do not edit by hand.

Common conventions:

- `comp` : composition name or numeric item id; omitted = the active composition.
- `layer` : layer name or 1-based index.
- `property` : a path such as `Transform/Position` , `Opacity` , `Effects/Gaussian Blur/Blurriness` , `Text/Animators/Animator 1` (names, matchNames or indices).
- Colors: `#RRGGBB` or `[r, g, b]` in 0..1 (0..255 is accepted too). Times are in seconds.
- Tools marked *one undo step* change the project and are undone with a single Ctrl+Z.



Property paths through a layer's property tree

Contents

- **Bridge and project:** `ae_health`, `ae_guidelines`, `ae_list_project`, `ae_open_project`, `ae_save_project`, `ae_get_result`
- **Scripting:** `ae_run_jsx`, `ae_run_jsx_file`
- **Looking at the result:** `ae_capture_frame`, `ae_capture_frames`, `ae_render_preview`, `ae_get_selection`, `ae_inspect`
- **Compositions and layers:** `ae_create_comp`, `ae_set_comp`, `ae_create_layer`, `ae_set_layer`, `ae_center_layers`, `ae_get_layer_timing`
- **Properties and animation:** `ae_set_property`, `ae_set_keyframes`, `ae_set_expression`, `ae_add_text_animator`, `ae_copy_animation`
- **Effects and presets:** `ae_list_available_effects`, `ae_apply_effect`, `ae_list_layer_effects`, `ae_remove_effects`, `ae_list_presets`, `ae_apply_preset`
- **Markers and audio:** `ae_add_markers`, `ae_get_audio_info`, `ae_set_audio_levels`, `ae_analyze_audio`

Bridge and project

`ae_health`

Check the After Effects bridge: server and panel versions, AE version, open project, active composition, panel status. Changes nothing.

No parameters.

ae_guidelines

Read the full motion-design and workflow guidelines for this bridge (easing, bounce, text animators, stagger, track mattes, motion blur, review loop, ExtendScript pitfalls). Call it once before designing or animating. Does not contact After Effects.

No parameters.

ae_list_project

List the open project: compositions with their layers (index, name, type, in/out points) and all other items (folders, footage, solids) with their folder and usedIn count (0 = unused). Changes nothing.

Parameter	Type	Description
<code>items</code>	boolean	Include non-composition items (default true).

ae_open_project

Open an .aep project in After Effects (closing the current one). If the current project has unsaved changes you must choose save: true or discard: true. Use this instead of opening/closing projects in `ae_run_jsx`, which breaks AE's undo history.

Parameter	Type	Description
<code>path</code> <i>(required)</i>	string	Absolute path to the .aep file.
<code>save</code>	boolean	Save the current project's changes before closing it.
<code>discard</code>	boolean	Close the current project without saving its changes.

ae_save_project

Save the open project. With no path saves to its current file; with an absolute .aep path saves there (Save As).

Parameter	Type	Description
<code>path</code>	string	Optional absolute .aep path.

ae_get_result

Fetch the result of a command that was still running when its tool call timed out. Never re-sends the command.

Parameter	Type	Description
<code>commandId</code> <i>(required)</i>	string	
<code>waitMs</code>	number	Wait up to this long for the result (default 0, max 600000).

Scripting

ae_run_jsx

Run arbitrary ExtendScript (ES3 JavaScript) inside After Effects. The value of the last expression is returned (AE objects come back as short summaries). Pass data through `args` (available as the `args` variable) instead of pasting it into the code. Helpers: `mcp.findComp(nameOrId?)`, `mcp.findLayer(comp, nameOrIndex)`, `mcp.activeComp()`, `mcp.layerType(layer)`, `mcp.projectInfo()`; `lib.prop(layer, 'Transform/Position')`, `lib.setValue(prop, value, time?)`, `lib.color('#ff8800')`, `lib.tree(propertyGroup, depth, withValues)`. The whole call is one undo step. If it runs longer than the timeout, a commandId is returned: fetch the result with `ae_get_result`, do not re-send the code.

Parameter	Type	Description
<code>code</code> <i>(required)</i>	string	ExtendScript code. ES3 only: var, no arrow functions, no let/const.
<code>args</code>	object	Optional data exposed to the code as <code>args</code> .
<code>timeoutMs</code>	number	How long to wait before returning a commandId (default 30000, max 3600000).

ae_run_jsx_file

Run an ExtendScript .jsx/.js file from disk (a library of reusable scripts) with optional `args`, like `ae_run_jsx`. One undo step.

Parameter	Type	Description
<code>path</code> <i>(required)</i>	string	Absolute path to the script.
<code>args</code>	object	
<code>timeoutMs</code>	number	

Looking at the result

ae_capture_frame

Render one frame of a composition to PNG and return it as an image for visual inspection. Transparent areas are filled with the composition background color by default. Changes nothing.

Parameter	Type	Description
<code>comp</code>	string number	Composition name or numeric item id. Omit to use the active composition.
<code>time</code>	number	Time in seconds. Omit to use the composition's current time.
<code>background</code>	string array	'comp' (default: composition background color), 'transparent', or a color '#RRGGBB' / [r,g,b].
<code>path</code>	string	Optional absolute PNG path to keep the file (saved as rendered). By default a temporary file is used and deleted.
<code>compareWith</code>	string	Optional absolute path to a reference image (PNG/JPEG/WebP/GIF), returned next to the frame for comparison.

ae_capture_frames

Render several frames of a composition into ONE contact-sheet image (grid with time labels) to judge an animation: timing, easing, overlaps, elements appearing outside their containers. Give explicit times, or start/end with count (default 9 evenly spaced) or step. Changes nothing.

Parameter	Type	Description
comp	string number	Composition name or numeric item id. Omit to use the active composition.
times	array	Explicit times in seconds.
start	number	Default 0.
end	number	Default: last frame of the comp.
count	number	Number of frames between start and end (default 9, max 36).
step	number	Seconds between frames, instead of count.
columns	number	Grid columns (default: square-ish).
maxWidth	number	Sheet width limit in pixels (default 2048).
background	string array	'comp' (default), 'transparent' (shown as gray) or a color.

ae_render_preview

Render a composition (or a time range) through the Render Queue to an H.264 MP4 or a PNG sequence for the user to watch. Other queued items are paused during the render and restored. Blocks After Effects while rendering; long renders return a commandId for ae_get_result.

Parameter	Type	Description
comp	string number	Composition name or numeric item id. Omit to use the active composition.
format	string: mp4 , png	Default mp4.
start	number	
duration	number	
half	boolean	Render at half resolution (faster).
path	string	Output file (mp4) or folder (png). Default: /renders.
timeoutMs	number	Default 600000.

ae_get_selection

What the user selected in After Effects: project items, layers of the active comp, and selected properties with their paths, values, keyframes (time, value, interpolation, ease) and property trees for selected groups (e.g. a text animator). Use it when the user says 'this', 'the selected layer', 'look at what I selected'. Changes nothing.

Parameter	Type	Description
depth	number	Tree depth for selected groups (default 3).

ae_inspect

Deep property tree of a layer or of one of its groups, with values, keyframes and easing, expressions. By default only modified properties are listed (modifiedOnly: false lists everything). Changes nothing.

Parameter	Type	Description
comp	string number	Composition name or numeric item id. Omit to use the active composition.
layer <i>(required)</i>	string number	Layer name or 1-based index.
property	string array	Optional group or property path to start from, e.g. 'Text/Animators/Animator 1'.
depth	number	Default 6.
modifiedOnly	boolean	Default true.
values	boolean	Default true.
keys	boolean	Include keyframe details (default true).
maxNodes	number	Default 800.

Compositions and layers

ae_create_comp

Create a composition (and open it in the viewer unless open: false). One undo step.

Parameter	Type	Description
name	string	
width	number	Default 1920.
height	number	Default 1080.
duration	number	Seconds, default 10.
frameRate	number	Default 30.
pixelAspect	number	Default 1.
bgColor	string array	Color: '#RRGGBB' or [r, g, b] in 0..1 (0..255 also accepted).
motionBlur	boolean	Composition motion blur switch.
open	boolean	

ae_set_comp

Change composition settings: name, size, duration, frame rate, background color, motion blur switch and shutter angle, work area, current time; open: true opens it in the viewer. One undo step.

Parameter	Type	Description
comp	string number	Composition name or numeric item id. Omit to use the active composition.
name	string	

Parameter	Type	Description
width	number	
height	number	
duration	number	
frameRate	number	
bgColor	string array	Color: '#RRGGBB' or [r, g, b] in 0..1 (0..255 also accepted).
motionBlur	boolean	Composition motion blur switch.
shutterAngle	number	
workAreaStart	number	
workAreaDuration	number	
time	number	Move the current-time indicator.
open	boolean	

ae_create_layer

Create a text, shape, solid, adjustment or null layer. Text is center-justified by default; use `ae_center_layers` to center any layer visually. One undo step.

Parameter	Type	Description
comp	string number	Composition name or numeric item id. Omit to use the active composition.
type (required)	string: text, shape, solid, adjustment, null	
name	string	
text	string	text: the text.
font	string	text: PostScript font name, e.g. 'Arial-BoldMT'.
fontSize	number	
justification	string: left, center, right	
color	string array	text fill / shape fill / solid color.
strokeColor	string array	Color: '#RRGGBB' or [r, g, b] in 0..1 (0..255 also accepted).
strokeWidth	number	
shape	string: rectangle, ellipse, polygon, star	shape: default rectangle.
size	array number	shape: [w, h] (default 200x200); solid/adjustment: [w, h] (default comp size).
roundness	number	shape rectangle corner roundness.
points	number	shape polygon/star points.

Parameter	Type	Description
<code>fill</code>	boolean	shape: add a fill (default true).
<code>position</code>	array	[x, y] in comp pixels. Default: center.
<code>startTime</code>	number	Time in seconds (composition time).
<code>duration</code>	number	Layer duration in seconds (default: to the end of the comp).

ae_set_layer

Change layer settings and transform: name, enabled, 3D, label, parent, timing, anchorPoint, position, scale, rotation, opacity, and text (text, font, fontSize, fillColor, strokeColor, strokeWidth, justification, tracking, leading). Pass time to write keyframes instead of static values. One undo step.

Parameter	Type	Description
<code>comp</code>	string number	Composition name or numeric item id. Omit to use the active composition.
<code>layer</code> (required)	string number	Layer name or 1-based index.
<code>name</code>	string	
<code>enabled</code>	boolean	
<code>threeD</code>	boolean	
<code>label</code>	number	Label color index 0..16.
<code>parent</code>	string number null	Parent layer, or null to unparent.
<code>startTime</code>	number	Time in seconds (composition time).
<code>inPoint</code>	number	Time in seconds (composition time).
<code>outPoint</code>	number	Time in seconds (composition time).
<code>anchorPoint</code>	array	
<code>position</code>	array	
<code>scale</code>	array number	Percent: 50 or [50, 50].
<code>rotation</code>	number	Degrees (Z rotation for 3D layers).
<code>opacity</code>	number	0..100.
<code>text</code>	string	
<code>font</code>	string	
<code>fontSize</code>	number	
<code>fillColor</code>	string array	Color: '#RRGGBB' or [r, g, b] in 0..1 (0..255 also accepted).
<code>strokeColor</code>	string array	Color: '#RRGGBB' or [r, g, b] in 0..1 (0..255 also accepted).
<code>strokeWidth</code>	number	
<code>justification</code>	string: <code>left</code> , <code>center</code> , <code>right</code>	

Parameter	Type	Description
<code>tracking</code>	number	
<code>leading</code>	number	
<code>motionBlur</code>	boolean	Layer motion blur switch; turning it on also enables the comp switch.
<code>trackMatte</code>	string number null	Matte layer (AE 2023+), or null to remove the track matte.
<code>trackMatteType</code>	string: <code>alpha</code> , <code>alphaInverted</code> , <code>luma</code> , <code>lumaInverted</code>	Default alpha.
<code>matteVisible</code>	boolean	Show (true) or hide the matte layer itself after assigning it.
<code>time</code>	number	Optional: set transform/text values as keyframes at this time.

ae_center_layers

Center layers in the composition. By default the anchor point is moved to the visual center of the layer content first (`sourceRectAtTime`), so text and shapes end up truly centered. One undo step.

Parameter	Type	Description
<code>comp</code>	string number	Composition name or numeric item id. Omit to use the active composition.
<code>layers</code>	array string number	Layer names/indices. Omit to use the selected layers.
<code>all</code>	boolean	Center every layer.
<code>axis</code>	string: <code>both</code> , <code>horizontal</code> , <code>vertical</code>	
<code>anchor</code>	boolean	Move the anchor point to the content center first (default true).

ae_get_layer_timing

Layer timing in seconds and frames: start, in/out points, duration, source in/out, stretch. Changes nothing.

Parameter	Type	Description
<code>comp</code>	string number	Composition name or numeric item id. Omit to use the active composition.
<code>layer</code> <i>(required)</i>	string number	Layer name or 1-based index.

Properties and animation

ae_set_property

Set any property of a layer or of its effects by path. With time, sets a keyframe; without time the property must have no keyframes. One undo step.

Parameter	Type	Description
<code>comp</code>	string number	Composition name or numeric item id. Omit to use the active composition.
<code>layer</code> (required)	string number	Layer name or 1-based index.
<code>property</code> (required)	string array	Property path: 'Transform/Position', 'Opacity', 'Source Text', 'Effects/Gaussian Blur/Blurriness' (names, matchNames or 1-based indices, '/'-separated or as an array).
<code>value</code> (required)	any	Number, array, color ('#RRGGBB' for color properties), text or text-style object for Source Text.
<code>time</code>	number	Time in seconds (composition time).

ae_set_keyframes

Add keyframes to a layer or effect property and set their interpolation. ease: linear | hold | bezier | easy (Easy Ease) | easyIn | easyOut; influence 0.1..100 (default 33.3); easeIn/easeOut: { speed, influence } for custom temporal ease; spatial props also accept spatialAutoBezier, spatialContinuous, roving, inTangent, outTangent. One undo step.

Parameter	Type	Description
<code>comp</code>	string number	Composition name or numeric item id. Omit to use the active composition.
<code>layer</code> (required)	string number	Layer name or 1-based index.
<code>property</code> (required)	string array	Property path: 'Transform/Position', 'Opacity', 'Source Text', 'Effects/Gaussian Blur/Blurriness' (names, matchNames or 1-based indices, '/'-separated or as an array).
<code>keys</code> (required)	array	[{ time, value, ease?, influence?, easeIn?, easeOut?, ... }]. Omit value to only change an existing key's ease.
<code>ease</code>	string: linear, hold, bezier, easy, easyIn, easyOut	Default ease for all keys.
<code>influence</code>	number	Default influence for all keys.
<code>replace</code>	boolean	Remove existing keyframes of this property first.

ae_set_expression

Set an expression on a layer or effect property, or remove it with an empty string. Reports expressionError if AE rejects it. One undo step.

Parameter	Type	Description
<code>comp</code>	string number	Composition name or numeric item id. Omit to use the active composition.
<code>layer</code> (required)	string number	Layer name or 1-based index.
<code>property</code> (required)	string array	Property path: 'Transform/Position', 'Opacity', 'Source Text', 'Effects/Gaussian Blur/Blurriness' (names, matchNames or 1-based indices, '/'-separated or as an array).

Parameter	Type	Description
<code>expression</code> (required)	string	

ae_add_text_animator

Add a text animator reveal (the motion-design way, instead of animating Transform): offsets such as position [0, 40], opacity 0, blur 16 are applied through a Range Selector that is animated away. shape 'rampUp' (default for characters/words) slides a soft window over the text so letters overlap in a wave; 'square' (default for lines) animates Start 0 -> 100 with strong ease. mode 'out' mirrors the reveal (last units leave first). Several layers can be staggered. One undo step.

Parameter	Type	Description
<code>comp</code>	string number	Composition name or numeric item id. Omit to use the active composition.
<code>layers</code>	array string number	Text layers. Omit to use the selected layers.
<code>offsets</code>	object	{ position, anchorPoint, scale, rotation, opacity, blur, tracking, fillColor, skew }. Default { position: [0, 40], opacity: 0, blur: 16 }.
<code>basedOn</code>	string: <code>characters</code> , <code>charactersExcludingSpaces</code> , <code>words</code> , <code>lines</code>	Default characters.
<code>shape</code>	string: <code>rampUp</code> , <code>square</code>	rampUp = overlapping wave, square = one unit after another.
<code>waveWidth</code>	number	rampUp: width of the soft window in % of the text (default 40; larger = more overlap).
<code>start</code>	number	Start time in seconds (default: layer in-point).
<code>duration</code>	number	Default 1.
<code>stagger</code>	number	Frames between layers (default 0).
<code>influence</code>	number	Ease influence of the selector keys (default 85 for square, 50 for rampUp).
<code>ease</code>	object	Range Selector Advanced ease: { high, low } in percent.
<code>mode</code>	string: <code>in</code> , <code>out</code>	
<code>name</code>	string	Animator name (default 'MCP Reveal').

ae_copy_animation

Copy a property or a whole property group with its keyframes, easing and expressions from one layer to other layers (like Ctrl+C / Ctrl+V), optionally staggered. Works for text animators ('Text/Animators/Animator 1'), effects ('Effects/Gaussian Blur'), masks and single properties ('Transform/Position'). Target n (1-based in the given order) is shifted by offset + n * stagger frames. One undo step.

Parameter	Type	Description
<code>comp</code>	string number	Composition name or numeric item id. Omit to use the active composition.
<code>from</code> (required)	string number	Source layer.

Parameter	Type	Description
property (required)	string array	Property or group to copy.
to	array string number	Target layers. Omit to use the selected layers.
stagger	number	Frames between consecutive targets (default 0).
offset	number	Extra time shift in seconds (default 0).
replace	boolean	For indexed groups (animators, effects): remove a same-named one on the target first.

Effects and presets

ae_list_available_effects

Search effects installed in After Effects by display name, matchName or category. Changes nothing.

Parameter	Type	Description
query	string	
max	number	Default 200.
includeObsolete	boolean	

ae_apply_effect

Apply an effect to a layer by display name or matchName and optionally set its properties. Returns the effect's properties so they can be adjusted next. One undo step.

Parameter	Type	Description
comp	string number	Composition name or numeric item id. Omit to use the active composition.
layer (required)	string number	Layer name or 1-based index.
effect (required)	string	e.g. 'Gaussian Blur' or 'ADBE Gaussian Blur 2'.
name	string	Optional new name for the effect instance.
settings	object	{ propertyNameOrPath: value }, e.g. { Blurriness: 20 }.

ae_list_layer_effects

List the effects on a layer with their property tree and current values. Changes nothing.

Parameter	Type	Description
comp	string number	Composition name or numeric item id. Omit to use the active composition.
layer (required)	string number	Layer name or 1-based index.
depth	number	Property tree depth (default 2).
values	boolean	Include values (default true).

ae_remove_effects

Remove effects from a layer by name, matchName or index (or an array of them), or all. One undo step.

Parameter	Type	Description
comp	string number	Composition name or numeric item id. Omit to use the active composition.
layer <i>(required)</i>	string number	Layer name or 1-based index.
effect	string number array	
all	boolean	

ae_list_presets

Find animation presets (.ffx) in the standard Adobe preset folders or given folders. Words in query must all match the preset path, e.g. 'blur in'. Changes nothing.

Parameter	Type	Description
query	string	
roots	array	Folders to search instead of the defaults.
max	number	Default 200.

ae_apply_preset

Apply an animation preset (.ffx) to a layer. Find presets with ae_list_presets. One undo step.

Parameter	Type	Description
comp	string number	Composition name or numeric item id. Omit to use the active composition.
layer <i>(required)</i>	string number	Layer name or 1-based index.
path <i>(required)</i>	string	Absolute path to the .ffx file.

Markers and audio

ae_add_markers

Add one or many markers to a layer, or to the composition when layer is omitted. One undo step.

Parameter	Type	Description
comp	string number	Composition name or numeric item id. Omit to use the active composition.
layer	string number	Layer name or index. Omit for composition markers.
markers <i>(required)</i>	array	[[time, comment?, duration?, label? (0..16), chapter?, url?]]

ae_get_audio_info

Audio info of a layer: source file path, audio levels and their keyframes, existing markers. Changes nothing.

Parameter	Type	Description
comp	string number	Composition name or numeric item id. Omit to use the active composition.
layer <i>(required)</i>	string number	Layer name or 1-based index.

ae_set_audio_levels

Set Audio Levels (dB) of a layer: static, at a time, or as a list of keyframes. One undo step.

Parameter	Type	Description
comp	string number	Composition name or numeric item id. Omit to use the active composition.
layer <i>(required)</i>	string number	Layer name or 1-based index.
level	number	dB for both channels.
left	number	
right	number	
time	number	Time in seconds (composition time).
keys	array	[[time, level] or [time, left, right]]

ae_analyze_audio

Analyze a WAV file: peak envelope and transient peaks (beats, hits). Pass path, or comp + layer to use the layer's source file (times are then converted to composition time). With addMarkers: true, places a marker at every peak (on that layer, or on the composition when only path is given; one undo step). Only uncompressed PCM/float WAV.

Parameter	Type	Description
path	string	Absolute path to a .wav file.
comp	string number	Composition name or numeric item id. Omit to use the active composition.
layer	string number	Layer name or 1-based index.
points	number	Envelope resolution (default 200, max 5000).
threshold	number	Peak threshold relative to the loudest point, 0..1 (default 0.6).
minGap	number	Minimum seconds between peaks (default 0.1).
addMarkers	boolean	
markerComment	string	Comment for created markers (default 'peak').
envelope	boolean	Include the envelope in the answer (default true).